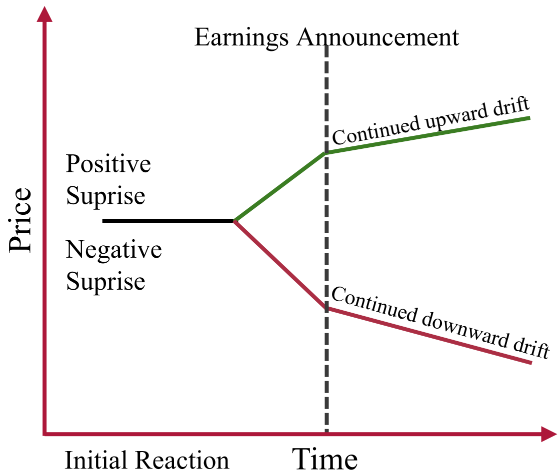
Triple-Filtered PEAD Strategy

Quant Group Pitch | 3/08/2026

PEAD Overview

Post-Earnings Announcement Drift (PEAD) refers to the tendency for stocks to continue moving in the direction of an earnings surprise for several weeks following the earnings release, typically lasting up to about 90 days. First documented in the seminal study The Post-Earnings Announcement Drift by Victor Bernard and Jacob Thomas in 1989, the phenomenon has since become one of the most widely studied anomalies in financial markets. Our research confirms that PEAD remains present in modern markets. However, the magnitude of the excess return has declined over time as the strategy has become more widely known and exploited. In the 2011-2025 sample period, the drift remains statistically observable, though the alpha decays more quickly and the overall return profile is weaker relative to earlier decades.



Methodology

To construct the research dataset, we compiled the full universe of companies in the S&P 500 and extracted earnings and price data covering the period from January 2011 to January 2026. This process produced approximately 28,000 earnings observations across 502 unique tickers. For each earnings announcement, we recorded the analyst consensus EPS estimate, the reported EPS, and the resulting earnings surprise. These earnings surprises were used to construct a Standardized Unexpected Earnings (SUE) measure. SUE scales the size of an earnings surprise by how variable that company's past surprises have been, making it possible to compare results consistently across very different companies. A firm that rarely beats estimates will have a high SUE even on a moderate beat, while a firm known for routinely large beats requires a bigger beat to score the same. Forward returns were calculated over three horizons: 21, 42, and 63 trading days, corresponding approximately to 30, 60, and 90 calendar days following the announcement. Cumulative Abnormal Returns (CARs) were constructed by subtracting the corresponding S&P 500 return over each window, removing broad market noise and isolating the post-earnings signal. Evaluating these horizons using cross-sectional averages and the Newey-West estimator (a statistical technique that accounts for the fact that overlapping holding periods create correlated data) revealed that the 21-trading-day window produced the strongest and most statistically significant drift, with a t-statistic of +4.01 and a p-value of 0.0002. The 42 and 63-day horizons produced progressively weaker results, consistent with prior academic literature showing that PEAD diminishes over time as information is gradually absorbed into prices. Accordingly, the 21-trading-day holding period was selected as the primary window for the strategy evaluation. A critical methodological priority was the elimination of look-ahead bias, which occurs when a model unintentionally uses information that would not have been available at the time a trade was made. Two specific issues were identified and corrected. First, SUE quintile rankings were recomputed using an expanding window approach: each company's ranking at the time of its announcement was calculated using only the data available up to that exact point, not the full quarter's worth of announcements. Second, earnings announcements made after the market's 4:00 PM close were shifted to enter at the following business day's open, since that information was not available to traders until after markets had closed. These corrections affected 34.8% of signals and changed the composition of the qualifying Q5 pool by approximately 1,400 signals compared to the naive stored values. Every trade in the corrected backtest could realistically have been executed using only information available at the time. Additionally, S&P 500 membership was enforced using point-in-time data, meaning a stock could only be traded in each period if it was in the index at that moment, eliminating survivorship bias from companies added retrospectively. Building on this foundation, the following sections outline the construction of a multi-factor trading strategy designed to isolate the most pronounced post-earnings anomalies, followed by a comprehensive statistical and stress-testing framework that includes Newey-West regression, bootstrap simulation, outlier fragility testing, crisis period analysis, and alpha decay evaluation.

Data Collection

- Universe: firms in the S&P 500
- Sample period: Jan 2011 - Jan 2026
- Data gathered: earnings announcements, consensus EPS, reported EPS, stock prices

Earnings Signal Construction

- Compute earnings surprise
- Standardize using historical volatility to create SUE
- Rank firms into SUE quintiles

Return Measurement

- Calculate forward returns after earnings over 21,42, 63 days
- Compute Cumulative Abnormal Returns(CAR)

Statistical Evaluation

- Compute cross-sectional average CAR
- Estimate Newey-West adjusted t-statistics
- Identify the 21-trading-day horizon as the strongest PEAD signal

Bias Corrections & Strategy Testing

- Remove look-ahead bias
- After-hours earnings: next-day trade entry
- Remove survivorship bias
- Robustness tests: Monte Carlo, outlier stress tests, crisis analysis

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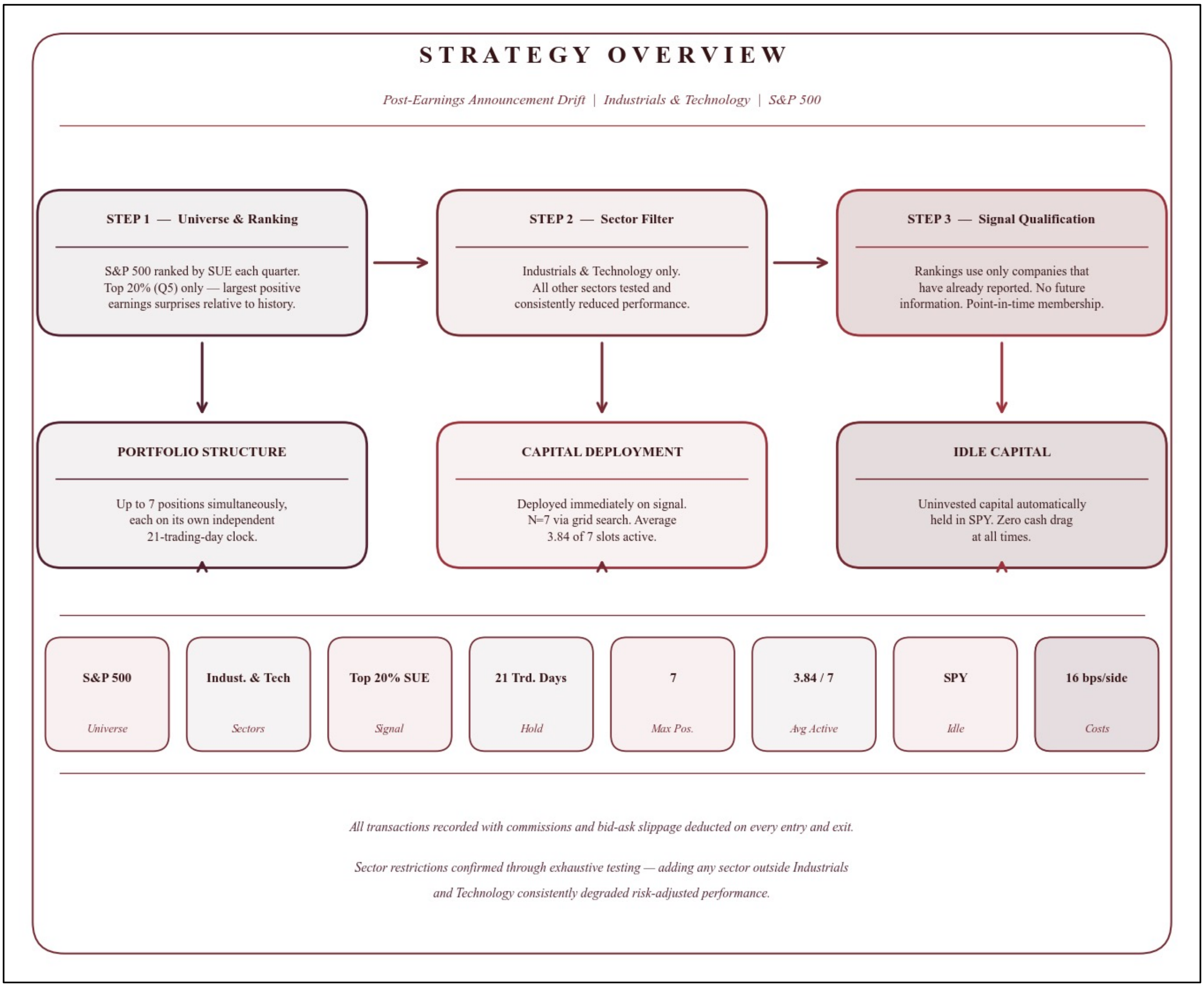
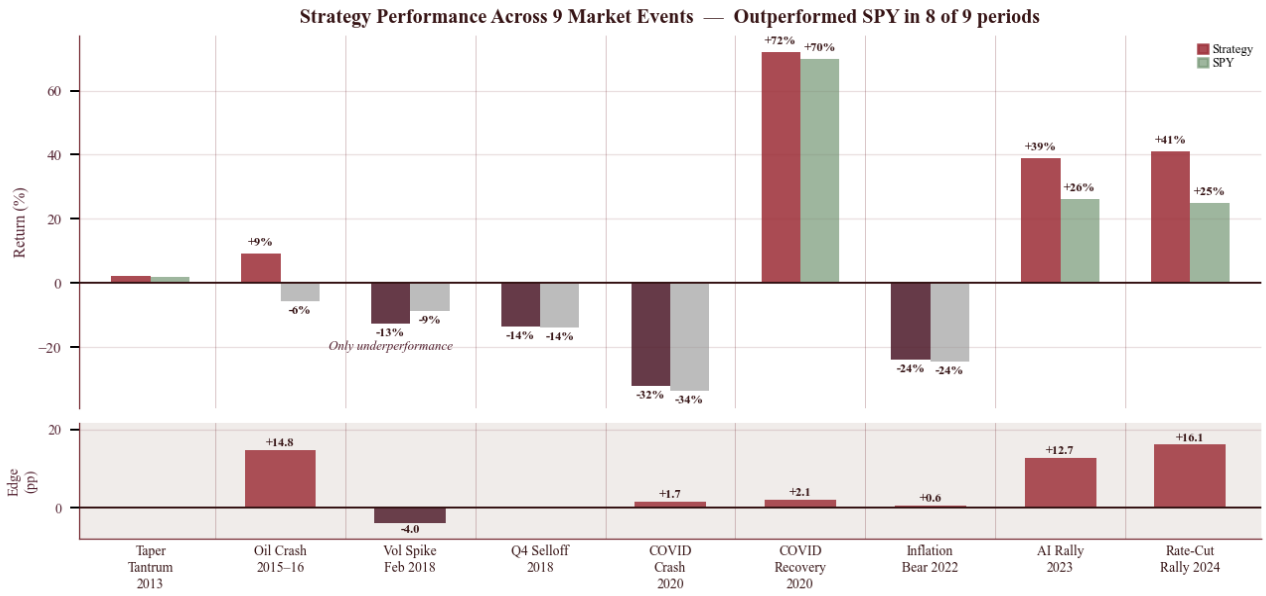
Strategy Overview

The trading strategy is designed to systematically capture the Post-Earnings Announcement Drift anomaly using a focused, rules-based framework that combines SUE magnitude, sector targeting, and dynamic capital management. Each calendar quarter, all companies within the S&P 500 are ranked by their Standardized Unexpected Earnings scores. The strategy focuses exclusively on the top 20% of these firms (Quintile 5 or Q5), representing companies that delivered the largest positive earnings surprises relative to their own historical volatility. Critically, these rankings are assigned using only the companies that have reported up to each given date, so no future information contaminates the signal. Since PEAD does not behave uniformly across the entire market, the strategy restricts eligible trades to two sectors: Industrials and Technology. Historical analysis of the full S&P 500 universe showed these two sectors produced the strongest, most statistically significant post-earnings drift. Adding other sectors, including Consumer Cyclical, Healthcare, and Financials, was tested exhaustively and consistently degraded both absolute returns and risk-adjusted performance, confirming that the edge is genuinely concentrated in these two sectors. When a qualifying signal is identified, capital is deployed immediately into that position regardless of what else is currently held, creating a rolling portfolio where positions are constantly entering and exiting on their own independent 21-day clocks. At any given moment, the portfolio holds between zero and seven active positions simultaneously, each started on a different date and each running its own 21-day countdown independently. The maximum of seven positions was selected through a systematic grid search, with N=7 producing the strongest risk-adjusted results given the available signal supply. On average 3.84 of the seven slots are active at any given time, reflecting the natural rhythm of earnings seasons. Any capital not deployed in an active position is automatically held in SPY, ensuring it is always earning a return rather than sitting idle. The backtest records every transaction and explicitly deducts estimated trading commissions and bid-ask spread slippage on every entry and exit.



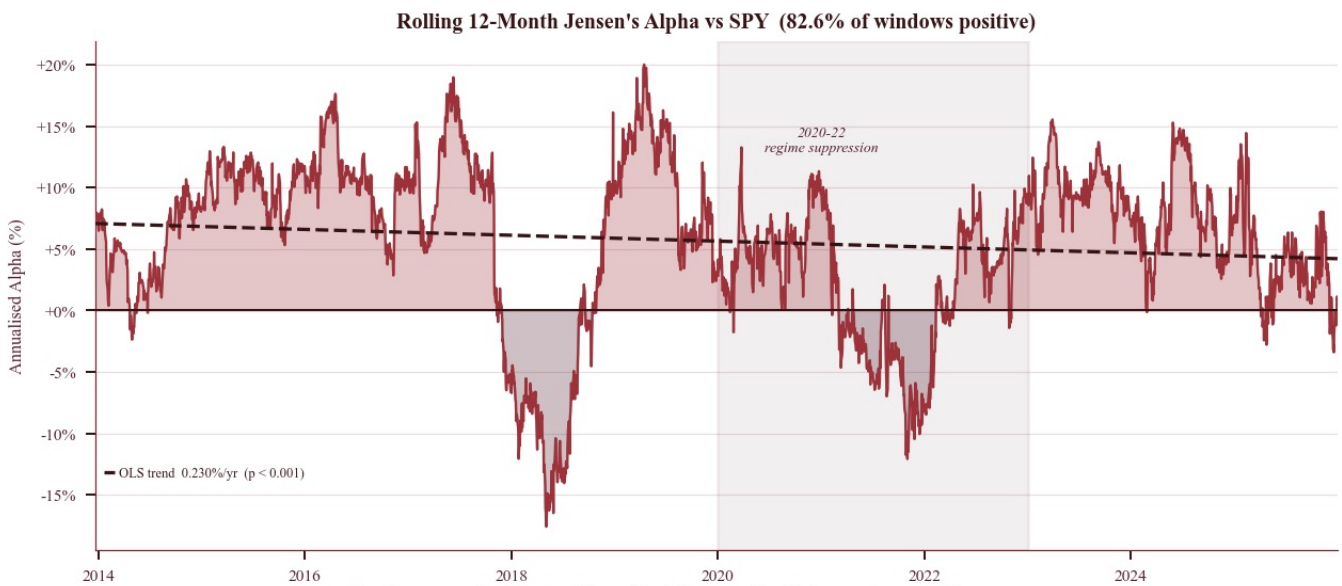
Strategy Results Statistical Tests

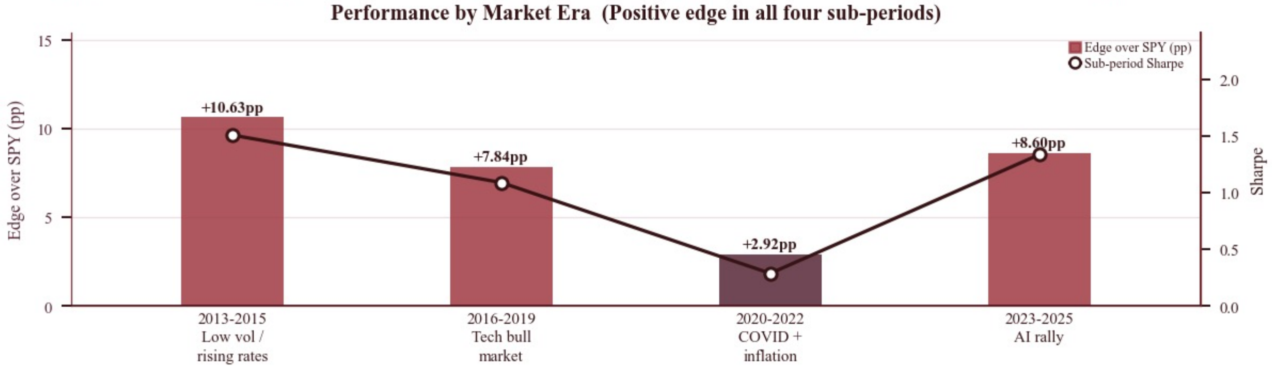
To confirm the strategy's outperformance is not a byproduct of data mining or random chance, the portfolio's value over time was subjected to an extensive statistical and stress-testing framework. The results are reported candidly, including areas of mixed evidence, as an honest assessment of the strategy is more useful than a selectively positive presentation. The foundational test, a one-sample t-test on the strategy's daily returns, yielded a t-statistic of +3.432 with a p-value of 0.000606, confirming a statistically significant positive return drift. The strategy's mean daily return of +0.087% compounds to the observed annual performance over the sample period. The shape of the return distribution shows a very mild negative skewness of -0.165, meaning the distribution is nearly symmetric, and a high excess kurtosis of 8.285. The slight negative skew is common for long-equity strategies during broad market drawdowns and at this magnitude it is not a meaningful concern. The return distribution has occasional large moves in both directions, but on 95% of trading days the portfolio loses no more than 2.02%, while the worst 5% of days average a loss of 3.16%, and even the most severe 1% of days average -5.19%, which is painful but recoverable given the strategy's overall annual return of around 21%. Risk-adjusted performance remains statistically significant when accounting for market exposure and serial correlation. A CAPM regression produced an annualized Jensen's alpha of +5.99% ($p = 0.0276$) with a market beta of 1.09, indicating that the strategy generates excess return beyond what would be expected from simple market exposure alone. Extending this analysis to Fama-French three-factor, five-factor, and momentum-augmented regressions produced nearly identical annual alphas of roughly +6.4% to +6.6%, all statistically significant, indicating that the strategy's returns are not explained by standard equity risk factors such as size, value, profitability, investment behavior, or momentum exposure. The Newey-West test, which accounts for multiple overlapping positions that can inflate statistical significance, still confirms the edge is real with a p-value of 0.0103. The strategy's Sharpe ratio of 0.936 is also statistically significant under the adjustment proposed by Andrew W. Lo (2002), reinforcing the conclusion that the observed risk-adjusted returns are unlikely to be explained by noise alone. To examine whether the results are sensitive to the specific historical ordering of trades, a Monte Carlo simulation generated 10,000 hypothetical portfolios by randomly drawing daily returns from the strategy's own return distribution. The randomized sequences produced a mean CAGR of +6.73% with a 95% confidence interval of +0.45% to +13.24%, far below the strategy's realized +21.65% CAGR, and only 1.3% of randomized sequences managed to beat the SPY benchmark, suggesting that the strategy's performance is strongly dependent on the actual timing of trades generated by the signal rather than merely the distribution of individual trade outcomes. An outlier fragility analysis examined how performance holds up as the best trades are systematically removed. After precisely removing the top 10% of all trades from the record, the strategy's estimated annual return remained at +21.60%, still comfortably above the SPY benchmark of +14.30%, confirming that the returns are not dependent on a handful of lucky outliers but are instead generated broadly across hundreds of ordinary trades. The strategy also demonstrated strong resilience to trading costs. Even when transaction costs were modeled at 100 basis points per side, more than six times the 16 basis points used in the base simulation and far above realistic execution costs for liquid S&P 500 stocks, the strategy still produced a +20.66% CAGR, maintaining a substantial performance advantage over the market. Historical stress testing across nine major market events further supports the robustness of the signal, with the strategy outperforming the benchmark during eight of the nine periods, including the 2015-2016 oil crash, where the strategy gained +9.09% while SPY fell -5.68%, and the 2023-2025 AI rally, where the strategy produced large excess returns during a strong bull market. The only period of underperformance occurred during the February 2018 volatility spike, which is consistent with a known vulnerability of momentum-based strategies during sudden volatility shocks and rapid factor reversals. During the 2020 COVID crash the strategy's maximum drawdown of -32.05% was like the market's -33.72% decline, indicating that extreme liquidity shocks temporarily overwhelm the earnings signal and cause the portfolio to track overall market behavior.



Strategy Alpha Decay Analysis

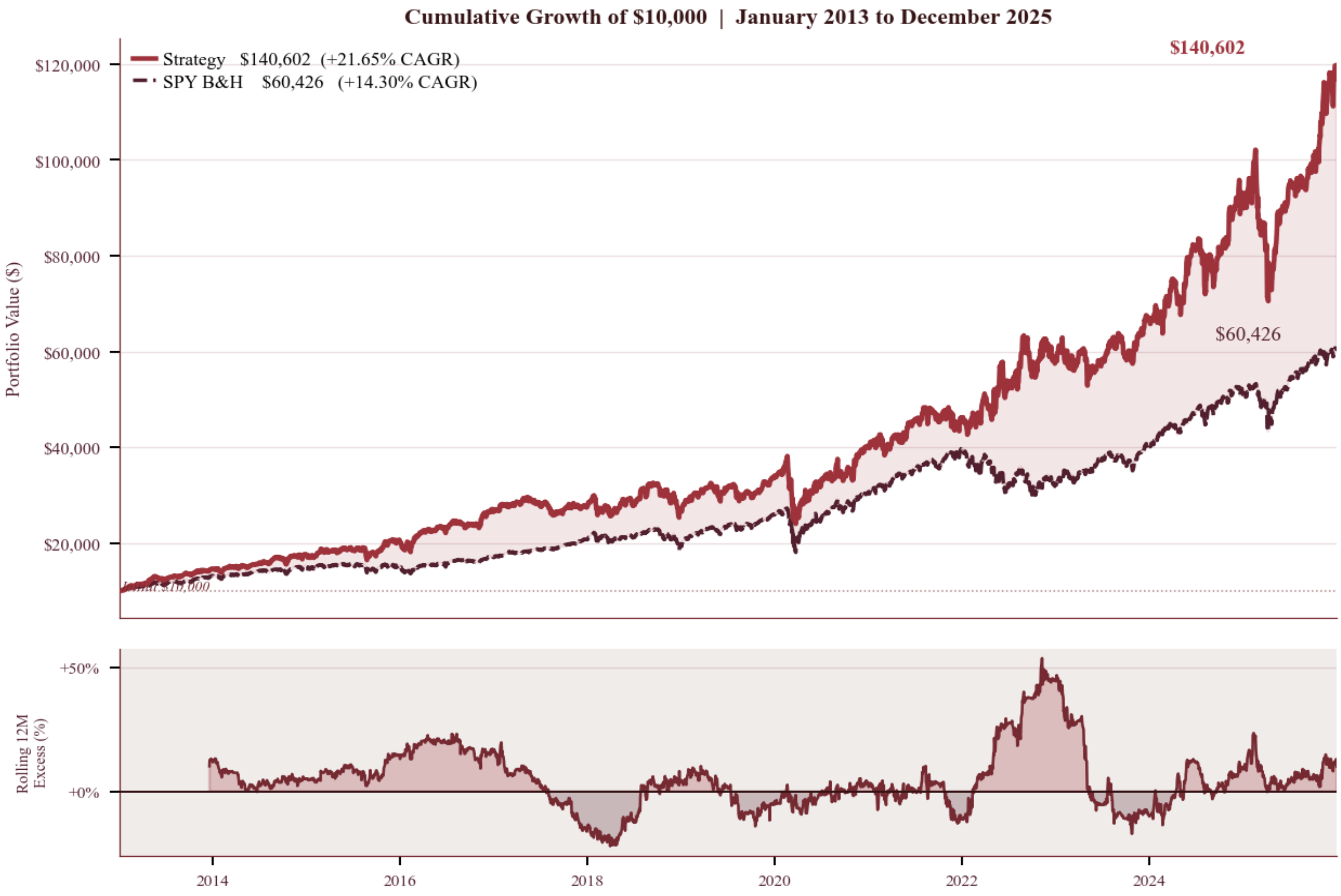
A key risk for any systematic strategy based on a known market anomaly is that the edge gradually disappears as more market participants become aware of and attempt to exploit it. To evaluate this risk, the strategy's alpha was analyzed through multiple sustainability tests across the full 12-year window. Using a CAPM regression on daily excess returns, the portfolio generates a statistically significant annualized Jensen's alpha of +5.99% over the SPY benchmark ($p = 0.0276$), representing the portion of return that cannot be explained simply by exposure to the overall market. The Information Ratio of +0.633 and Appraisal Ratio of +0.539 further confirm that the strategy produces consistent risk-adjusted excess returns relative to the benchmark. A rolling 12-month alpha analysis shows that 82.6% of all windows produced a positive alpha, meaning the strategy outperformed the market on a risk-adjusted basis during most periods. However, the OLS trend line fitted to the rolling alpha series exhibits a statistically significant downward slope of −0.230% per year, and a Mann-Kendall non-parametric trend test confirms this direction with a z-statistic of −8.918 ($p < 0.001$). At face value, these results suggest a gradual decline in the magnitude of the strategy's edge over time. A closer look at the sub-period results provides important context. Dividing the sample into four distinct market regimes shows that the strategy delivered annualized edges of +10.63 percentage points over SPY in 2013-2015, +7.84 points in 2016-2019, +2.92 points in 2020-2022, and +8.60 points in 2023-2025. The apparent downward trend in the rolling alpha is therefore largely driven by the 2020–2022 period, which coincided with three extraordinary macro environments: the COVID liquidity crash, the post-COVID speculative growth and meme-stock phase, and the 2022 inflation-driven bear market. Each of these episodes temporarily weakened the earnings signal as broader economic forces drove all stocks up or down together, regardless of individual company results. Performance during the 2023-2025 AI-driven rally suggests that the signal remains economically relevant when markets return to a more typical fundamental environment, with the strategy achieving a sub-period Sharpe ratio of 1.338, the strongest of the four eras. To test whether the strategy fundamentally changed after the COVID period, a Chow structural break test split the full history in January 2020 and compared the two halves. The result ($p = 0.749$) confirms the pre and post-COVID periods are statistically indistinguishable from each other, meaning the strategy did not change character after that point The average monthly outperformance over SPY dipped slightly from +0.67% before 2020 to +0.50% after, but this difference is too small to be statistically meaningful and is consistent with normal variation. The temporary weakness in the strategy during that period reflects the unusual market conditions of 2020 to 2022 rather than a permanent weakening of the underlying edge. Overall, the strategy demonstrates positive excess returns in all four sub-periods, which is an important structural robustness test. A strategy that produces consistent alpha across multiple market regimes is generally more credible than one whose historical performance is driven primarily by a single favorable environment.





Strategy Returns

The complete strategy was backtested over the 12-year period from January 2013 through December 2025 with all look-ahead safeguards applied throughout the pipeline. SUE quintile rankings were constructed using an expanding-window methodology to ensure that only information available at the time of each earnings announcement was used, and all after-close announcements were shifted to the following trading day's open to prevent forward bias. Over this horizon, the strategy delivered a cumulative return of +1,306%, corresponding to an annualized return (CAGR) of +21.65%. This substantially outperformed the S&P 500's +14.30% annualized return, representing an annualized edge of +7.34%. Starting from a $10,000 initial investment, the strategy grew to $140,602 compared with $60,426 for a passive SPY buy-and-hold allocation. After accounting for $15,270 in total transaction costs, the strategy still produced approximately $64,900 of additional profit relative to SPY, demonstrating that the edge remains economically meaningful even after realistic trading frictions. For additional perspective, an investor contributing $833 per month ($10,000 annually) over the same horizon would have accumulated $666,843 using the strategy versus $378,920 with SPY, creating approximately $287,924 in additional wealth from the identical contribution schedule. Risk-adjusted performance was strong across standard measures. The strategy achieved a Sharpe Ratio of 0.936, indicating substantial return per unit of total volatility, and an Information Ratio of 0.633, which is generally considered strong for an active strategy evaluated over a multi-year window. The maximum peak-to-trough decline during the sample was -32.05%, comparable to the S&P 500's drawdown of roughly -33%, despite the strategy generating significantly higher cumulative returns. A CAPM regression confirms that the outperformance is not simply the result of amplified market exposure. The strategy exhibits a market beta of 1.087, meaning it tends to move slightly more than the market during broad swings, but it also generates a statistically significant annualized Jensen's alpha of +5.99% (p = 0.0276) that cannot be explained by this market exposure alone. The R-squared of 0.725 indicates the strategy moves broadly in line with the market, which is expected since it is always at least partially invested in SPY. The remaining 27.5% of daily returns that cannot be explained by general market movements represents the contribution of the earnings selection process and it is precisely that 27.5% that generates the +5.82% annual alpha on top of market returns. Statistical significance was verified through multiple independent tests. Together, these results suggest that the observed performance is unlikely to be explained by random variation. Performance was also consistent at the annual level. The strategy outperformed the S&P 500 in 10 of 12 calendar years, with 11 of the 12 years producing a positive absolute return. The primary year of underperformance was 2021, when returns were dominated by a narrow mega-cap technology rally, and 2025, where results were broadly similar to the market.



Benchmark Comparison and Sector-Adjusted Performance

Because the strategy trades exclusively within the Technology and Industrials sectors, a comparison against the broad S&P 500 alone overstates the degree of genuine alpha. A passive investor seeking exposure to the same sectors could simply hold a blend of QQQ and XLI rather than running an active strategy. To address this, performance was evaluated against three benchmarks of increasing relevance: SPY as the broad market baseline, QQQ as the dominant technology proxy, and a 70% QQQ / 30% XLI blend as the sector-matched passive alternative that most closely approximates the strategy's investable universe. Six metrics are used throughout this section to evaluate performance on a risk-adjusted basis. The Sharpe ratio measures annualized return above the risk-free rate per unit of total volatility, where higher values indicate more return earned per unit of risk taken. The Information Ratio is the equivalent of Sharpe but with a benchmark substituted as the hurdle rather than cash, measuring how consistently the strategy outperformed that benchmark after accounting for the volatility of that excess return, values above 0.5 are generally considered strong. Jensen's alpha is the portion of return that cannot be explained by the strategy's exposure to the benchmark, tested for statistical significance. The Treynor ratio measures annualized excess return per unit of systematic beta rather than total volatility, making it more appropriate when the strategy is held as part of a broader portfolio. M-squared rescales the strategy to the benchmark's volatility and asks what return it would have delivered on a like-for-like risk footing, expressed as a percentage difference. The Omega ratio is the ratio of all probability-weighted gains to all probability-weighted losses above zero, capturing skew and tail behavior that Sharpe cannot. Upside and downside capture measure what fraction of the benchmark's positive and negative days the strategy participated in, where a ratio above 100% on the upside and below 100% on the downside represents the ideal profile.

SPY (Baseline) Comparison

Against SPY, the strategy delivered an annualized edge of +7.34 percentage points, a Sharpe ratio of 0.936 versus SPY's 0.767, an Information Ratio of 0.633, and a statistically significant Jensen's alpha of +5.82% per year ($p = 0.032$). The Omega ratio of 1.210 confirms the distribution of returns is net positive in a way that accounts for the full shape of the return distribution, and M-squared of +2.80% indicates the strategy would have delivered 2.80 percentage points more than SPY if both were normalized to identical volatility. Upside capture of 113.9% and downside capture of 107.0% reflect the higher-beta profile of 1.087, with the strategy participating more fully in both market rises and declines.

70% QQQ / 30% XLI Comparison

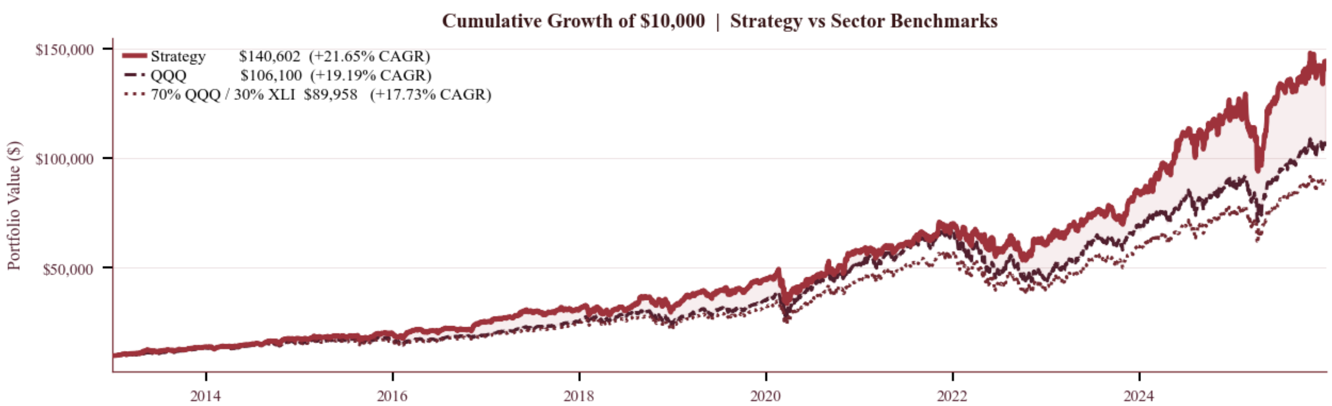
Against the 70% QQQ / 30% XLI sector-matched blend, which returned +17.73% annually with a Sharpe ratio of 0.843, the strategy outperformed by +3.91 percentage points per year with an Information Ratio of 0.331 and a Jensen's alpha of +4.71% ($p = 0.077$) that, while narrowly missing the 5% significance threshold, is consistent in direction and magnitude across all three benchmark comparisons. M-squared of +1.78% confirms the outperformance holds on equal-risk footing, and upside and downside capture of 97.2% and 92.3% respectively indicate the strategy participates in nearly all of the passive sector return while absorbing marginally less of the decline.

QQQ Comparison

Against QQQ, which returned +19.19% annually, the strategy outpaced it by +2.46 percentage points per year with a Sharpe ratio of 0.936 versus QQQ's 0.862. The Information Ratio of 0.181 is moderate, reflecting that this edge, while positive, is less consistent than the edge over SPY. The beta to QQQ of 0.862 means the strategy carries somewhat less Nasdaq concentration than a direct holding, and its Treynor ratio of 0.2304 exceeds QQQ's implied ratio. The upside capture of 87.5% and downside capture of 82.5% present a favourable asymmetry, the strategy participates proportionally less in QQQ's drawdowns than in its rallies.

Summary

Taken together, these results support the interpretation that the strategy's excess returns are not simply a repackaging of sector beta. An investor holding the sector-matched blend passively would have captured the same underlying exposure at lower cost and complexity yet would have given up nearly 4% of annual return. The strategy's ability to outperform a sector-matched benchmark consistently across twelve years through a rules-based earnings signal supports the conclusion that the alpha is structural rather than incidental.



Risks, Limitation, Caveats

While the strategy demonstrates robust historical outperformance across multiple independent statistical frameworks, its real-world application is subject to several structural risks and methodological limitations that any practitioner should carefully consider. The most acute risk is vulnerability to pure liquidity shocks. During events like the March 2020 COVID-19 crash, market-wide correlations converged toward 1.0 as forced liquidations and algorithmic de-risking dominated price action regardless of individual company earnings quality. In these environments, the strategy's fundamental signal temporarily loses all predictive value, and the portfolio tracks the broader market decline. The -32.05% drawdown during March 2020 reflects this directly. This is a structural limitation of all earnings-based strategies and cannot be eliminated without hedging instruments that would reduce returns in normal environments. By restricting eligible trades to the Industrials and Technology sectors, the portfolio intentionally excludes large segments of the modern economy including Consumer Discretionary, Healthcare, and Financials. While this restriction was validated empirically as improving signal quality, it introduces meaningful tracking error relative to the S&P 500 index and makes the strategy's relative performance sensitive to sector leadership in any given year. In narrow technology mega-cap rallies, where index returns are driven by a handful of the largest companies in sectors the strategy does not trade, the portfolio will naturally underperform on a relative basis even while generating strong absolute returns. Execution friction and position management present practical challenges in live trading. The backtest models 16 basis points of slippage per trade, a defensible estimate for liquid S&P 500 names under normal conditions. Live execution during volatile earnings announcements may encounter significantly wider bid-ask spreads, and market orders at the open on the morning after an earnings release carry higher execution uncertainty than mid-day trades. At capital sizes above approximately $100,000 per individual position, order sizes may begin to affect prices in thinner names, warranting a higher slippage assumption. The overlapping 21-day holding periods also require careful position management in live implementation to ensure new signals can be entered without disrupting existing positions. The look-ahead biases identified and corrected in this study, specifically the full-quarter SUE quintile assignment and the same-day entry on after-close announcements, were present in the original version of the backtest and have been fully addressed. Any live implementation must replicate these corrections precisely: quintile rankings must be computed using only companies that have reported up to the current date, and after-close announcements must be entered at the following day's open. Failure to do so would materially overstate expected performance, and a systematic process for computing real-time expanding-window quintiles is a prerequisite for live trading. Finally, while the alpha decay analysis found no evidence of structural degradation and the edge strengthened in the most recent period, the exploitable window for Post-Earnings Announcement Drift continues to compress over the long term as electronic markets become more efficient. High-frequency trading and machine learning-based strategies process earnings releases increasingly quickly, potentially reducing the duration over which the delayed price adjustment occurs. The rolling alpha and Information Ratio metrics described in the alpha decay section should be monitored continuously in any live deployment to provide early warning if this compression begins to affect the strategy's performance.

Key Risks and Limitations

Liquidity Shocks — HIGH
Market correlations converge to 1.0 during forced liquidations. Signal loses predictive value.

Sector Concentration — MEDIUM
Industrials & Technology only. Vulnerable to narrow mega-cap rallies in excluded sectors.

Execution Friction — MEDIUM
Wider spreads at earnings open. Order impact above ~$100k per position.

Look-Ahead Replication — HIGH
Live system must compute expanding-window quintiles in real time. Non-trivial.

Alpha Compression — LOW
HFT and ML strategies reduce drift window over time. Monitor IR continuously.

Fee & Slippage Sensitivity